MESIN KONVERSI ENERGI

ACUAN CEPAT • A5 • Rumus ideal: cek asumsi dan basis efisiensi

01 PISTON & BAHAN BAKAR

$V_{d,total} = z\pi D^2 L/4$; $r = (V_d,sil+V_c)/V_c$.
a = 2: 4 langkah; a = 1: 2 langkah.
$P_i = IMEP \cdot V_{d,total} \cdot N/(60a)$.
$P_\beta = 2\pi N\tau/60$; $P_{fr} = P_i - P_\beta$.
$\eta_m = P_\beta/P_i$; $BMEP = \eta_m \cdot IMEP$.
$\eta_b = P_\beta/(\dot{m}_f \cdot LHV)$; $\eta_b = \eta_m \cdot \eta_i$.
$\dot{m}_f = P_\beta/(\eta_b \cdot LHV)$ [kg/s].
$BSFC = 3{,}6\times10^6/(\eta_b \cdot LHV)$ [g/kWh].
LHV: kJ/kg; N: rpm; τ: N·m.
MEP: kPa + m^3 → kW; torsi → W.
$\eta_{comb} \neq \eta_i$; $\eta_b = \eta_m \cdot \eta_{comb} \cdot \eta_{conv}$.

02 DASAR & PILIH RUMUS

$Q - W = \Delta U$; $pv = RT$; $k = c_p/c_v$.
$\Delta u = c_v\Delta T$; $\Delta h = c_p\Delta T$; $R = c_p - c_v$.
Udara: R≈0,287; c_p≈1,005; c_v≈0,718 [kJ/(kg·K)], k≈1,4 (pendekatan).
Isentropik: pv^k=C; Tv^(k−1)=C.
$T_2/T_1 = (p_2/p_1)^{((k-1)/k)}$.
v tetap: w=0; $q = c_v\Delta T$.
p tetap: $w = R\Delta T$; $q = c_p\Delta T$.
Adiabatik tertutup: $w = c_v(T_{in} - T_{out})$.
$pv^n = C$: $w = (p_2v_2 - p_1v_1)/(1-n)$, $n \neq 1$.
Aliran tunak adiabatik: $w_{out} = h_{in} - h_{out}$.

03 OTTO & DIESEL (k KONSTAN)

Urutan 1→2→3→4→1; $r = v_1/v_2$.
Keduanya: 1–2 & 3–4 isentropik;
4–1 buang kalor pada v tetap.
Otto: 2–3 pada v tetap.
$T_2 = T_1 r^{(k-1)}$; $T_4 = T_3/r^{(k-1)}$.
$q_{in} = c_v(T_3 - T_2)$; $q_{out} = c_v(T_4 - T_1)$.
$\eta_O = 1 - 1/r^{(k-1)}$.
Diesel: 2–3 pada p tetap; $\rho = v_3/v_2$.
$T_3 = \rho T_2$; $T_4 = T_3(\rho/r)^{(k-1)}$.
$q_{in} = c_p(T_3 - T_2)$; $q_{out} = c_v(T_4 - T_1)$.
$\eta_D = 1 - (\rho^k - 1)/[k(\rho-1)r^{(k-1)}]$.
$w_{net} = q_{in} - q_{out}$; $\eta = w_{net}/q_{in}$.

04 BRAYTON & BOILER

1→2 kompresor; 3→4 turbin.
$w_c = h_2 - h_1$; $w_t = h_3 - h_4$; $q_{in} = h_3 - h_2$.
$w_{net} = w_t - w_c$; $\eta = w_{net}/q_{in}$.
BWR=wc/wt; $P_{net} = \dot{m} \cdot w_{net}$.
$\dot{m}_{air} = p_1\dot{V}_1/(RT_1)$; $rp = p_2/p_1$.
$\eta_{Brayton,ideal} = 1 - 1/rp^{((k-1)/k)}$.
$\eta_c = (h_2s - h_1)/(h_2a - h_1)$.
$\eta_t = (h_3 - h_4a)/(h_3 - h_4s)$.
Boiler: $\dot{Q}s = \dot{m}s(h_{out} - h_{in})$.
$\dot{m}_f = \dot{Q}s/(\eta_{boiler} \cdot LHV)$.
HR=3600/ηfuel→elec [kJ/kWh].
$\eta_{fuel\to elec} = \eta_{boiler} \cdot \eta_{cycle} \cdot \eta_{gen}$.*

05 RANKINE / ECE

1→pompa→2→boiler→3→turbin→4 →kondensor→1; ideal: $s_2 = s_1$, $s_4 = s_3$.
$w_p = h_2 - h_1 \approx v_1(p_2 - p_1)$; $w_t = h_3 - h_4$.
$q_{in} = h_3 - h_2$; $\eta = (w_t - w_p)/q_{in}$.
$P_{net} = \dot{m}(w_t - w_p)$; BWR=wp/wt.
$\eta_p = (h_2s - h_1)/(h_2a - h_1)$.
$\eta_t = (h_3 - h_4a)/(h_3 - h_4s)$.
Jenuh: h=hf+xhfg; x=(s−sf)/sfg.
x=0 cair; x=1 uap kering; 0≤x≤1.
Perbaikan: pcond↓; Tturbin,in↑; pboiler↑; reheat; regenerasi.
Reheat: turbin HP→panas→turbin LP.

06 TEORI & CEK CEPAT

ICE: bakar internal, gas bekerja langsung.
ECE: panas eksternal → fluida terpisah.
HHV/GCV > LHV/NCV: HHV termasuk kalor kondensasi air produk pembakaran.
SI knock: end-gas autoignition;
CI knock: delay → kenaikan p cepat.
Oktan↑: tahan knock SI; cetane↑: lebih mudah menyala / delay CI↓.
η_b vs rpm: sering optimum di tengah; kurva tergantung kondisi beban.
$\eta_m = P_\beta/P_i \leq 1$; $\eta_{th} = w_{net}/q_{in} \leq 1$.
a=aktual; s=isentropik; $\dot{m}$=kg/s.

SATUAN & BATAS PEMAKAIAN

1 bar=100 kPa; 1 cc=10^{-6} m^3; 1 kWh=3600 kJ; 1 hp≈0,7457 kW; T[K]=T[°C]+273,15.
1 ton/jam=1000/3600 kg/s; MJ/kg ×1000 → kJ/kg; kcal/kg ×4,184 → kJ/kg (ikuti soal).
*η_{cycle} berbasis kalor ke fluida. Jika η_{th} sudah berbasis fuel, jangan kalikan η_{boiler} lagi.
Q masuk (+), W keluar (+); q,w per kg. η_i & η_b adalah efisiensi termal berbasis fuel.
Gas ideal & kalor jenis konstan hanya bila sesuai; uap gunakan tabel. Cetak 100% / actual size.